PRODUCT DEMONSTRATION GUIDE

Developer: [Left Blank] | OptiCrop Project Documentation
Repository: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

1. Demo Flow and Presentation Script

This guide outlines the steps to demonstrate OptiCrop features in a live presentation:

- Scenario A (High Yield Case): Input optimal parameters for rice (N=90, P=40, K=40, temp=24, rainfall=220) to show the crop suggestion and yield forecast.
- Scenario B (Nutrient Deficit Case): Input low Nitrogen values (N=10) to show the NPK engine recommending Urea application.
- Scenario C (Diagnostics): Show the K-Means clustering category description.
- Scenario D (Analytics): Review the feature rankings and confusion matrix on the analytics tab.

Opticrop - Demo Script

Segment 1 - Input Form (2 min)
Show form fields: N, P, K, pH, rainfall, temperature, humidity.
Demonstrate out-of-bounds validation error.

Segment 2 - Prediction Results (3 min)
Submit valid inputs -> display Crop suggestion, Yield (t/ha),
Soil cluster, NPK correction table.

Segment 3 - Analytics Dashboard (3 min)
Accuracy comparison chart, confusion matrix, feature importances.

Segment 4 - Unit Test Run (2 min)
Run 'python -m pytest tests/' live; show 4/4 passed.

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